FACULTY OF COMPUTING
UTM Johor Bahru

SCST1223: PROBABILITY & STATISTICAL DATA ANALYSIS
2025/2026 – SEMESTER 2

PROJECT Phase 2

In this Project phase 2, you and team members are required to conduct an inference statistical analysis that cover the following analysis based on the collected dataset in phase 1.

- ✓ Compulsory:
  - Hypothesis testing 1-sample or 2-sample
  - Correlation
  - Regression
- ✓ Optional (at least 1):
  - Goodness of fit test
  - Chi Square test of independence
  - ANOVA

The findings are represented in report (refer Project Report.pdf for the format) and video. The video is suggested to upload in LinkedIn with a short caption/description on what's the project about, and the findings. Do tag the members as well. The video shall be publicly access and the URL shall be provided as appendix in the report. Details of submission as follows:

- ✓ One submission folder will be created in e-learning for you to upload a **ZIP file** containing supporting files (**19 JUNE 2026, Before 11.59 P.M)**
  - *R* Script files
  - Dataset
  - Project Report.pdf

The due date for this project is **WEEK 14.**

REPORT FORMAT

Times New Roman, size 12 and spacing 1.15.
One column (normal report).
5-8 pages (not including Appendix).

REPORT ORGANIZATION

1) **Introduction or background**
   Explain the purpose of the study in your own words, why you were interested in the topic, what you want to investigate and what you expected to see from the data.

2) **Dataset**
   Description of the collected dataset including parameters or variable to be measured and how you do the data pre-processing / filtering (if applicable). Also, explanation on why you choose these parameter/ variables and possible outcome of the test based on your chosen variables.

3) **Data analysis**
   - 3 compulsory tests:
     i. Hypothesis 1 or 2 sample test
     ii. Correlation test
     iii. Regression test

   - ONE optional test (at least). Choose from:
     i. Goodness of fit test
     ii. Chi-Square test of independence
     iii. ANOVA test

   For each test, you need to define the variables used and what is your conclusion from the test.

   For Correlation and Regression, you MUST include scatter plot graph to represent the analysis. Interpret and discuss what you can learn from the tests and graph.

***NOTE: PLEASE DO NOT INCLUDE R SCRIPT CODE IN YOUR REPORT.**
The script is submitted in another R or text file.

4) **Conclusion**
   What have you learned from all activities done in Project 2 (dataset, pre-processing and analysis process etc.) What is your best/interesting findings from your results?

5) **Appendix**
   A single page reflection about using Dr Quant to facilitate in the learning process.
   Screenshot on the conversation specifically on the output.
   Sample of your original/raw dataset and your processed dataset (if applicable).
   Video URL